



Internship Report

Prediction of RNA 3D structure using gradient-descending and knowledge-based scoring function

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Abstract

Write an abstract here.

Key words: financial models, machine learning...

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1 Presentation of the company

2 Preliminary notions

The goal of this chapter is to give insights on preliminary notions that are important to comprehend my report.

2.1 Central Dogma

Every cells of every living being, excluding virus and a few exceptions, has **DesoxyriboNucleic Acid (DNA)**. This molecule contains the genetic information. It holds all information for an organism. However the DNA alone cannot do anything, it needs to be express in proteins. This process is called the central dogma of molecular biology. For that, the cell perform two steps :

1. Transcription : This steps copy a part of DNA and transcribes it into a intermediate molecule called **RiboNucleic Acid (RNA)**, or more precisely messenger RNA (mRNA). The transcription is performed inside the nucleus.
2. Translation : The RNA obtained in the previous step are now used to make a protein. The translation is performed outside of the nucleus.

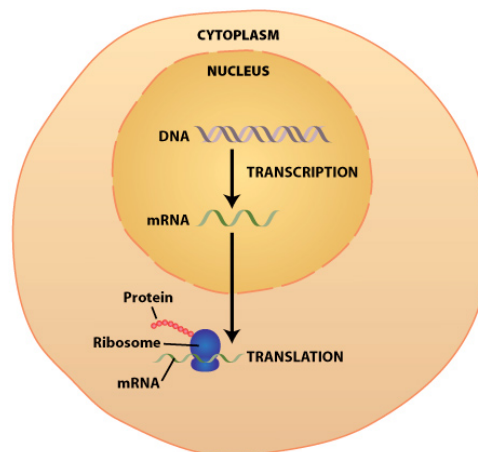


Figure 1: A diagram illustrating the processes of DNA transcription into RNA and RNA translation into proteins. Source : [1]

Recently, researchers have discovered that RNA serves more than just an intermediate role. Indeed, many RNAs, known as non-coding RNAs, are never translated into proteins. Instead, they have a regulation role inside the cell.

2.2 RNA molecule

The RNA is single-stranded, unlike DNA which is double-stranded, and it's a polymer of nucleotids. The linear arrangement of these nucleotids defines it's primary structure (1D structure) or it's raw sequence. A nucleotide is composed by three components (Fig. 2) : a nucleobase, a ribose sugar and a phosphate group. Unlike phosphate group and ribose sugar who is common for all nucleotids, the nucleobase determine the type of nucleotide. For the RNA, there are four types of nucleobases : Guanine (G), Adenine (A), Cytosine (C) and Uracil (U).

Structure of RNA

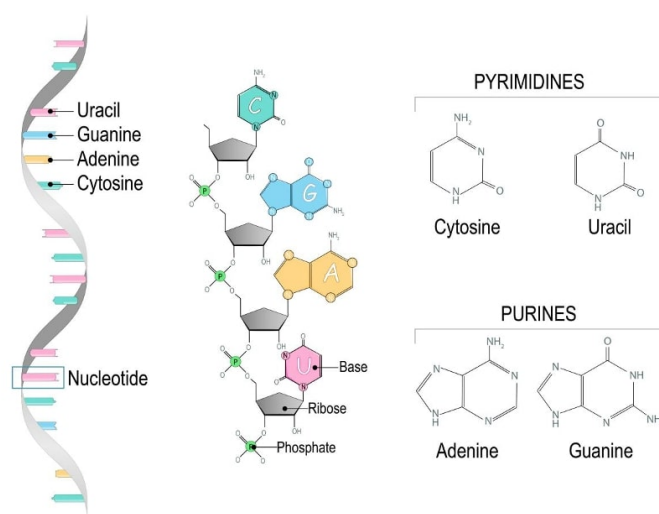


Figure 2: A diagram illustrating the 1D-structure of RNA and nucleotid composition. Source : [3]

There is two form of nucleobase : pyrimidines for uracil and cytosine, purines for adenine and guanine. Purines have a second ring, so they have more atoms and are bigger than pyrimidines. These nucleobase can assemble by stacking on top of each other trough van der Waals interactions. There is also structural constraint influences by ribose sugar chain, this chain is called the backbone [7]. And finally, the phosphate group, carry negative energy so to neutralize this charges, RNA needs heteroatoms like ions or ligand to regulate it's energy. So the conformation of RNA it's very dependant of the environnement condition, it's surrounding solvant and ions has a strong influences on it's stability and it's folding [5].

2.3 RNA secondary structure

The secondary or 2D structure is dictated by the primary structure. The 2D structure define the canonical interactions, such as Watson-Crick base pairing (A paring with U and G paring with C) and Wobble pair (G with U). The secondary structure alone can't define the folding of a RNA. Indeed, many pair can give multiple possible arrangement in 3D space (see Fig. 3). 2D structure can also be driven by the RNA functions and interactions with proteins or other molecules. Like for 1D structure, the secondary structure influence completely the 3D structure, so knowing the RNA secondary structure for make a prediction can be effective and having a better accuracy.

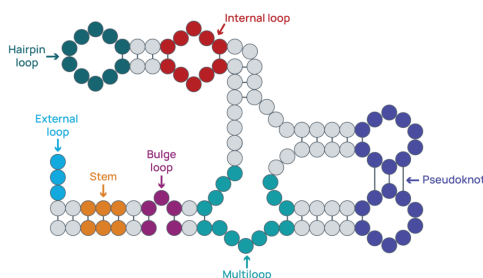


Figure 3: A diagram illustrating different structural motifs . Source : [6]

2.4 RNA tertiary structure

The transition from the secondary structure to the tertiary structure, or 3D structure, is determined by the geometric and topological properties of the motifs formed in the previous step. These motifs such as loops or bulges act as flexible hinges and anchor points.

Thanks to the flexibility of these nodes, the strand continues to fold. Long-range interactions then arise between 2D motifs that may be very far apart along the 1D structure but are geometrically close to one another in space. For example, a loop located at one end may interlock with the cavity of another region.

It is this dense network of interactions between the different motifs that forces the RNA to fold in a highly specific manner to adopt its final three-dimensional structure. In biology, a fundamental principle dictates that structure determines function. The tertiary structure allow the RNA molecule to interact with other element from the cell.

So to know the folding and the function of a RNA we need to know the 1D structure (4a). This will be used to determine the secondary structure (4b), to finally predict the 3D structure (4c).

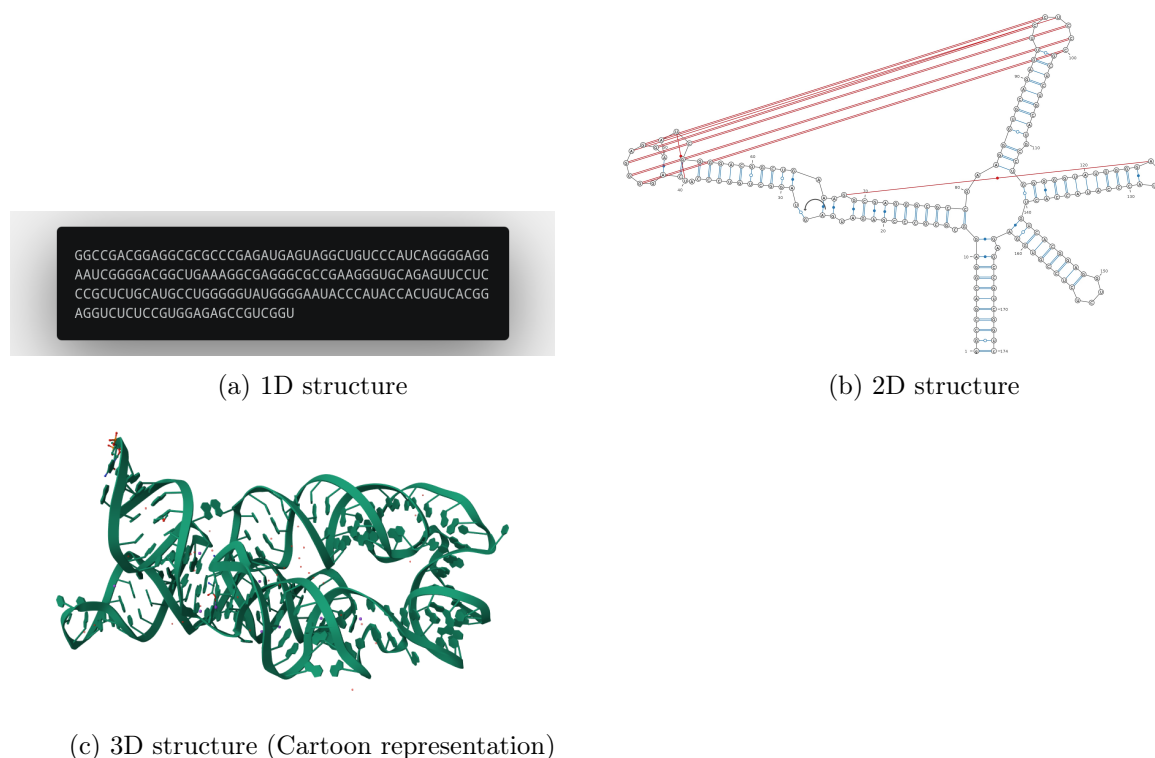


Figure 4: The three different level of representation for the same RNA (PDB ID : 3DIQ)

3 Introduction

4 Data and Dataset

4.1 PDB database

The **Protein Data Bank (PDB)** [2, 8] is the principal worldwide archive for three-dimensional structures of biological macromolecule such as proteins and nucleic acids. This database centralize all structure determined by experimental methods such as X-ray crystallography, nuclear magnetic resonance spectroscopy and cryo-electron microscopy. It can be accessed through its website and an API. Each structure has an identifier composed of four letters or numbers (e.g. 1J8G) and they can be download in two format : PDB file and **macromolecular Crystallographic Information File (mmCIF)** files. A format change happen in 2014, with the advent of cryo-electron microscopy, researchers had the tools to see larger molecules like RNA polymerase. These types of molecules have a lot of atoms but PDB file are restricted to only 99 999 atoms, with mmCIF files, there is no limit on the molecule's scale. As a result, every new molecule added to the database is now in this format. These two format will be describe in the next section.

4.2 PDB file structure

The PDB file is a file to represent each atom from a molecule, especially for protein and RNA. Each line have a keywords to explicite the line content. The file can be split in 3 parts.

The first one contains header and metadata. This part have all information about the molecule and the experimentation.

Keywords	Description
HEADER	Molecule Type, PDB Id, deposition date
TITLE	Title of the experiment
COMPND	Description of the molecule compenent
SOURCE	Biological origin or Synthetic
KEYWDS	Descriptive keywords for the structure
EXPDTA	Experimental Methods
AUTHOR	Author of the deposit
REVDAT	Revision history
JRNL	Publication reference
REMARK	Comments numbered from most important to least important

Table 1: All keywords founded in a first part of a PDB files

The table 1 present all keywords and his signification. REMARK contains useful information such as the resolution of the molecule. The resolution is the difference between the atoms found experimentally and the actual atoms. More the resolution is low, more the experimental molecules can be accurate.

The second part are about the sequence and the crystallography parameters. These keywords are all in the table 2

Keywords	Description
SEQRES	Primary Sequence by chain
CRYST1	Crystal lattice parameters
SCAL1/2/3	Orthogonal coordinate conversion matrices

Table 2: All keywords founded in a second part of a PDB files

A chain in an alternative conformation for the same molecule, sometimes by adding some ligand or other molecules to induce the change of conformation. SEQRES can be used to find

the acid nucleics sequence that can be retrieved and put inside a algorithm to predict the three-dimensional structure.

The last part is the goal of this file, the coordinates of every atoms from the molecule. This section is formatted in fixed columns. It's not just generic space between columns, the position is important. The information for every line is resume in the table 3.

```
ATOM      1  O5'   U A      1      12.967  20.885  19.962  1.00  5.19           0
```

Figure 5: Line 808 from the pdb file from the pdb id 1J8G [4]

Columns	Fields	Value (from Fig. 5)	Description
1-6	Record Name	ATOM	Two types: ATOM for standard residues and HETATM for heteroatoms
7-11	Serial Number	1	Sequential atom number
13-16	Atom Name	O5'	Atom name per IUPAC nomenclature
17	AltLoc	blank	Alternate location indicator, used when an atom has more than one modeled conformation (A/B)
18-20	Residue Name	U	Residue type (e.g. uracil for RNA)
22	Chain ID	A	Chain identifier
23-26	Residue Sequence Number	1	Residue position within the chain
31-38	x	12.967	Cartesian X coordinate (Å)
39-46	y	20.885	Cartesian Y coordinate (Å)
47-54	z	19.962	Cartesian Z coordinate (Å)
55-60	Occupancy	1.00	Fraction of the unit cell in which this atom occupies this position (1.00 = always present)
61-66	Temperature Factor	5.19	Thermal displacement / positional uncertainty (B-factor); lower = sharper, more ordered atom
77-78	Element	O	Chemical element symbol

Table 3: All keywords found in the coordinate section of a PDB file

4.3 Difference between PDB file and mmCIF file

PDB files and mmCIF files have the same goal: describe the structural content. They have the same type of information, coordinates, experimental metadata. For the same PDB id, these two files must describe exactly the same molecule. The difference between them is not about the information but about the structure file.

A PDB file uses fixed-width columns, so a parser needs to know exactly the specification. On the contrary, mmCIF is a key-value format, where each value is explicitly named. This format is more general and can be used by more tools and more easily. Because each value is a free token instead of a fixed-width column, mmCIF has no limit on chain ID length or residue number, which is why it can scale to every molecule, even massive assemblies like ribosomes or cryo-EM structures that a PDB file cannot represent. On the other hand, a mmCIF file is less user-friendly when you want to read information without a parser. These differences between these file types explain why both extensions are usually available inside the PDB database, except for very large structures where only mmCIF can be provided.

5 Materials and Methods

5.1 Knowledge based score function

5.2 Objective function

5.3 Optimization algorithm

5.4 Complete Pipeline

5.5 Different Applications

6 Results

6.1 Interpolation results

6.2 Comparaison between original and predicted structures

6.3 Comparatives tables

6.4 Line chart of score function for folding trajectory

6.5 Internet platform for RNA folding

7 Discussion and Perspectives

8 Conclusion

Glossary

D | **M** | **P** | **R**

D

DNA DesoxyriboNucleic Acid. [1](#),

M

mmCIF macromolecular Crystallographic Information File. [5](#),

P

PDB Protein Data Bank. [5](#),

R

RASP Ribonucleic Acids Statistical Potential.

RNA RiboNucleic Acid. [1](#),

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